

 OR-AS Operations Research Applications and Solutions	Case Name: Building Nevele	Sector	Construction (residential building)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
Submitted by	Stef Pauwels and Emile Van de Walle		One of nine std. scenarios
Date	June 2023		User defined distributions
File Name	C2023-07	Project Control	Automatic tracking
			Tracking based on user input

1. Project description

Project authenticity

This project consisted of constructing an apartment building three stories high. The construction was relatively straightforward for the company. The building was built on an empty piece of land and the employed techniques were all well known throughout the company. The construction consisted of individual activities that are relatively standard in this industry.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	42
Planned Duration (PD)	481 days
Budget At Completion (BAC)	€743 676
Renewable Resources	-
Consumable Resources	-

* standard eight-hour working days

Network topology	
Serial/Parallel (SP)	47%
Activity Distribution (AD)	48%
Length of Arcs (LA)	46%
Topological Float (TF)	23%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0	0	N/A
CRI-rho	100	0	N/A
CRI-tau	100	0	N/A

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	0.49	0.44	0.14
SI	0.34	0.37	0.6
SSI	0.12	0.12	1.05
CRI-r	0.13	0.11	1.34
CRI-rho	0.12	0.1	1.45
CRI-tau	0.08	0.07	1.74

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	9.95	-2.12	1	0	5.83
PV - SPI	11.86	5.83	CPI	0	5.83
PV - SCI	11.86	5.83	SPI	7.81	-0.43
ED - 1	11.53	-0.43	SPI(t)	8.62	5.83
ED - SPI	11.86	5.83	SCI	7.81	5.83
ED - SCI	11.86	5.83	SCI(t)	8.62	-3.58
ES - 1	6.76	-3.58	0.8 CPI + 0.2 SPI	1.91	5.17
ES - SPI(t)	9.36	5.17	0.8 CPI + 0.2 SPI(t)	2.15	5.17
ES - SCI(t)	9.36	5.17			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 28 tracking periods with a length of approximately one week. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	0	1	0	1	1	1	1
std dev	N/A	N/A	N/A	N/A	N/A	N/A	N/A
final	0	1	0	1	1	1	1

2.3.3.2. Time forecasting

PD	481 days
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Real Duration	352 days
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Late	26.8%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	348.36	N/A	-0.21	0.21
PV - SPI	348.36	N/A	-0.21	0.21
PV - SCI	348.36	N/A	-0.21	0.21
ED - 1	348.36	N/A	-0.21	0.21
ED - SPI	348.36	N/A	-0.21	0.21
ED - SCI	348.36	N/A	-0.21	0.21
ES - 1	348.36	N/A	-0.21	0.21
ES - SPI(t)	348.36	N/A	-0.21	0.21
ES - SCI(t)	348.36	N/A	-0.21	0.21

2.3.3.3. Cost forecasting

BAC	€743 676
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Real Cost	€743 676
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On budget	0
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	743676	N/A	0	0
CPI	743676	N/A	0	0
SPI	743676	N/A	0	0
SPI(t)	743676	N/A	0	0
SCI	743676	N/A	0	0
SCI(t)	743676	N/A	0	0
0.8 CPI + 0.2 SPI	743676	N/A	0	0
0.8 CPI + 0.2 SPI(t)	743676	N/A	0	0